

PowerFlow

Unlocking the Dual Nature of LLMs via Principled Distribution Matching

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CODE

01 THE GAP IN RLIF

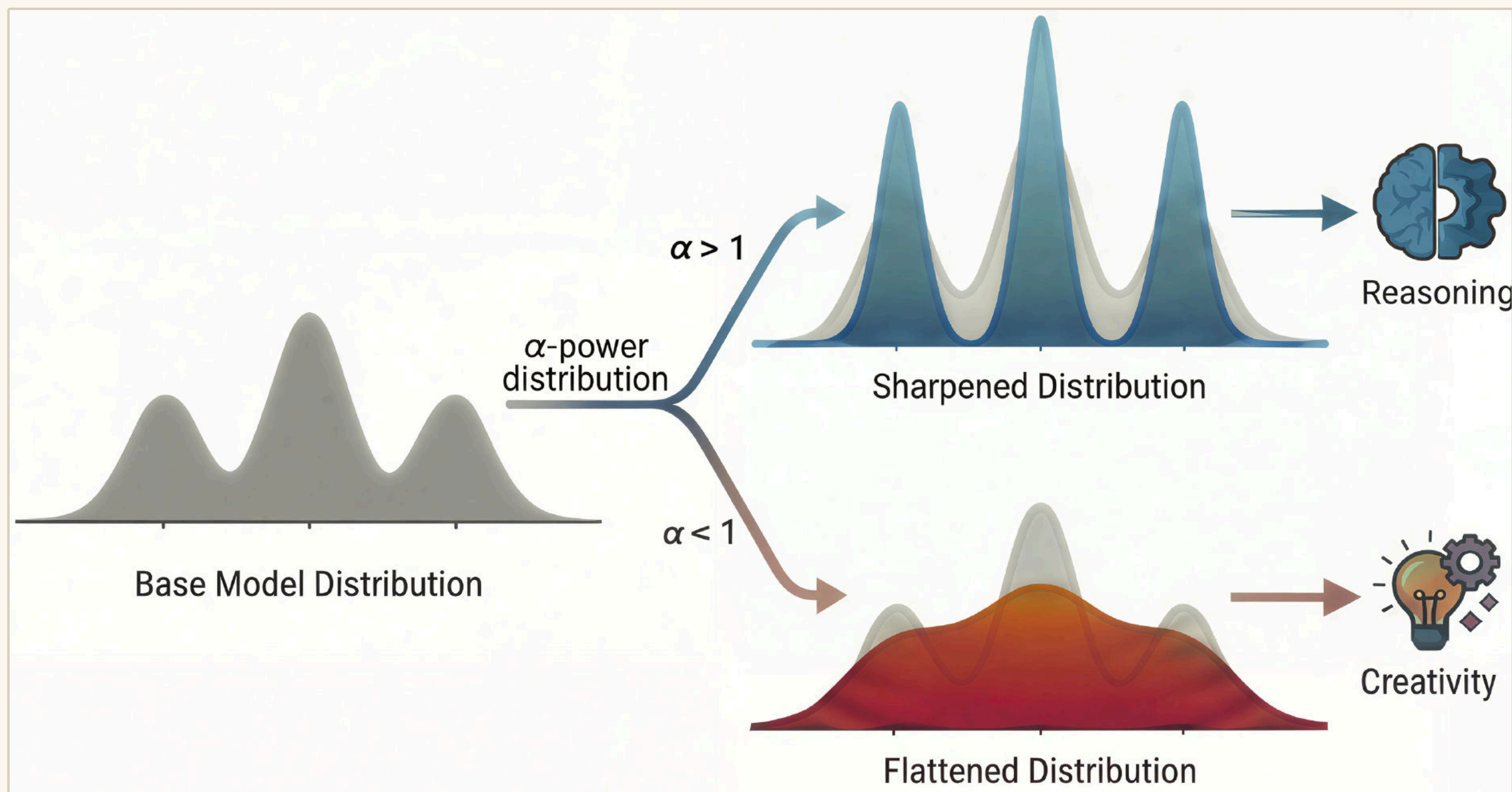
Unsupervised **RLIF** elicits latent ability with no labels — yet most methods lean on a *hand-designed intrinsic reward*. Lacking a principled optimization target, these heuristics drift into pathological behaviour during over-optimization:

- **Length collapse or explosion** of outputs.
- **Overconfidence** from driving entropy down for its own sake.
- **Mode collapse** onto one strategy, erasing diversity.

Q. Can one *principled* target replace hand-crafted rewards — and steer a model toward reasoning or creativity on demand?

02 TWO NATURES, ONE EXPONENT

PowerFlow matches the **α -power (escort) distribution** of the base model, $p_\alpha(y | q) \propto p_{\text{base}}(y | q)^\alpha$ — reshaping it by one exponent while **preserving relative rankings and mode structure** (the length-aware objective relaxes this across lengths). The exponent α is a single knob.



The dual nature. One base distribution → sharpened ($\alpha > 1$, reasoning) or flattened ($\alpha < 1$, creativity), set by a single exponent (flatten → creativity applies to aligned models).

$\alpha = 4$ (base) for reasoning · $\alpha = 0.5$ for creativity.

03 MATCHING, NOT REWARDING

Casting **GFlowNet** as an amortized variational sampler turns distribution matching into an on-policy objective. A **length-aware Trajectory-Balance** (LA-TB) loss projects it onto a length-normalized energy surface, neutralizing the exponential length bias of autoregressive generation.

LENGTH-AWARE OBJECTIVE

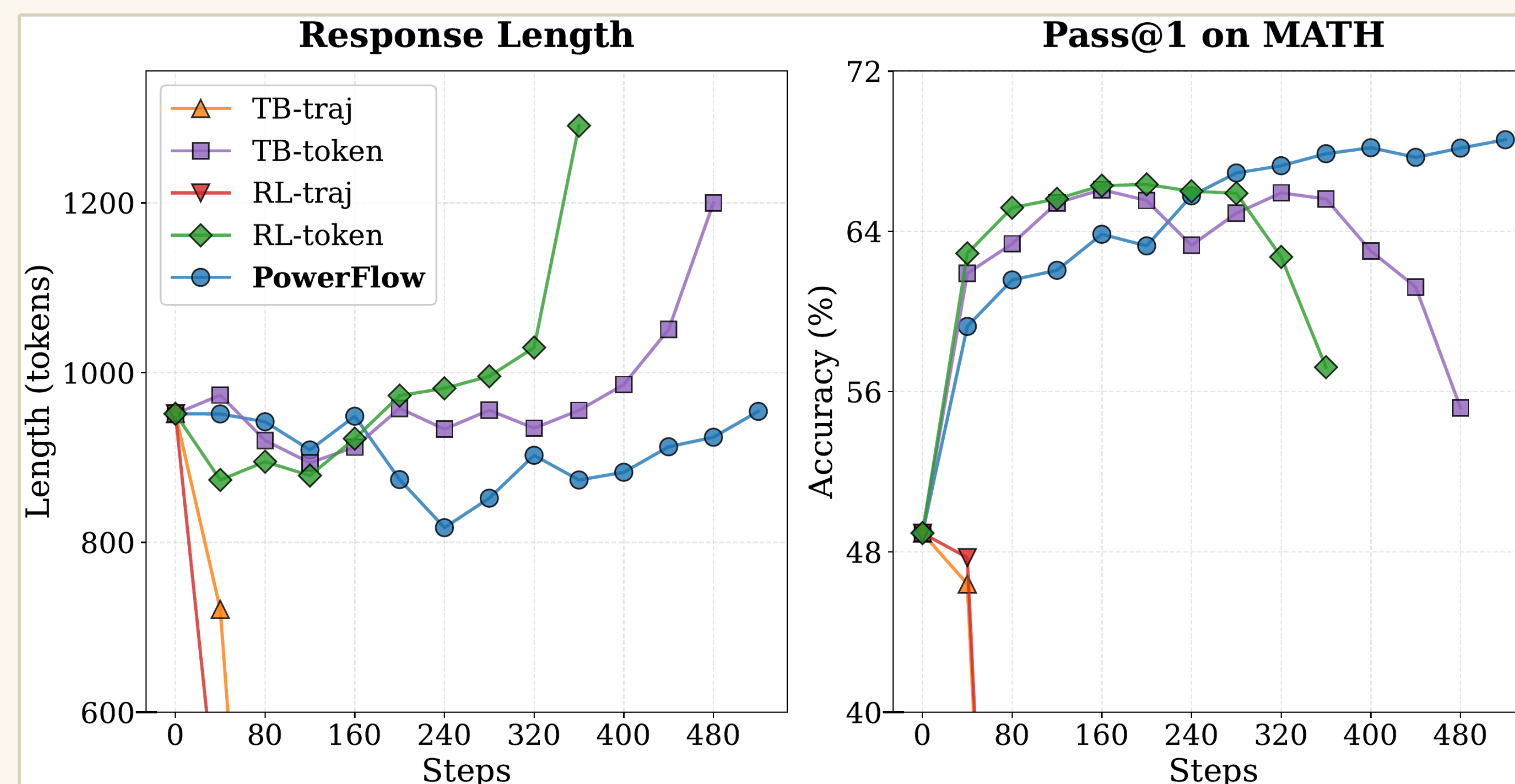
$$\mathcal{L}_{\text{LA-TB}} = \left(\log Z'_\phi(q) + \frac{1}{|y|} \log \frac{\pi_\theta(y | q)}{\tilde{p}_{\text{target}}(y | q)} \right)^2$$

No reward model, no correctness verifier, no labels — the target $\tilde{p}_{\text{target}} = p_{\text{base}}^\alpha$ with a light format penalty.

One knob. Reasoning sweep $\alpha \in \{2, 4, 6\}$ lands on $\alpha = 4$ (base; $\alpha = 2$ for aligned) — the same knob at $\alpha = 0.5$ restores creativity.

04 STABLE BY DESIGN

Trajectory-level matching (TB / RL-traj) collapses response length almost immediately; token-level variants inflate likelihood with repetition, then decay. PowerFlow holds length stable and accuracy on an overall upward trend — the length-aware objective is what makes matching usable on LLMs.



Training dynamics. Only PowerFlow keeps length stable, with accuracy staying on an overall upward trend on MATH.

05 SHARPEN → REASONING

Qwen2.5-Math-7B · avg@16 · sharpening $\alpha = 4$

Method	MATH500	AIME25	AMC23	GPQA	Avg*
Base	46.7	4.2	34.5	29.7	24.95
EMPO	79.3	12.3	60.2	36.0	40.88
TTRL	80.4	11.9	58.8	34.7	41.18
PowerFlow	78.1	14.4	63.4	37.0	42.17
GRPO†	78.4	12.9	63.4	34.4	42.38

42.17

best unsupervised avg

3 / 4

configs beat GRPO

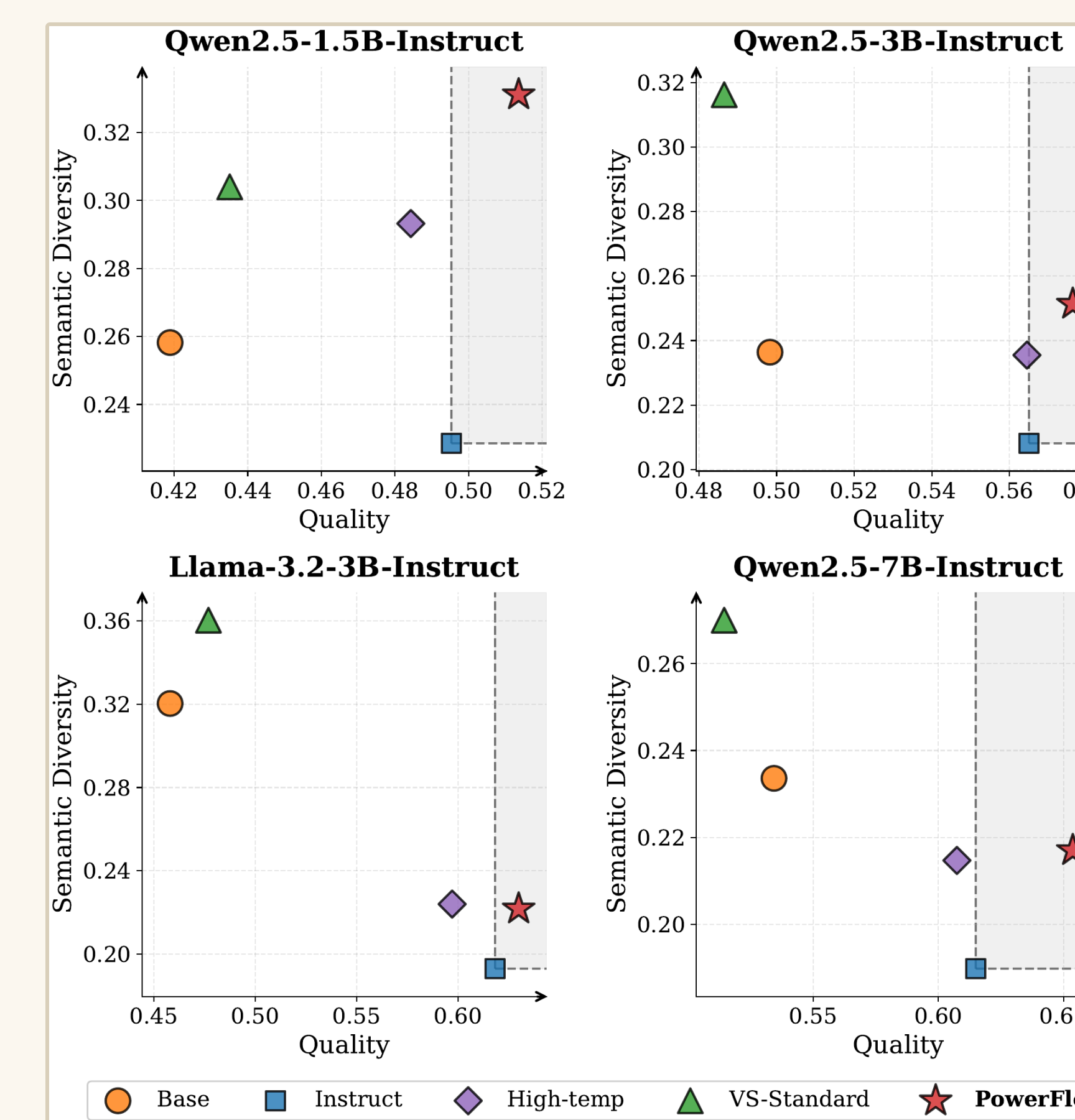
4.05

diversity > GRPO 3.93

Best 6-benchmark average among unsupervised baselines, matching supervised †GRPO (42.17 vs 42.38). *Avg = mean of six benchmark avg@16 scores (Olympiad, AIME24 omitted). Gold = best unsupervised.

06 FLATTEN → CREATIVITY

Flattening aligned models ($\alpha = 0.5$) counteracts alignment's *typicality bias*: PowerFlow **uniquely** raises quality *while* lifting semantic diversity.



Quality vs. semantic diversity. PowerFlow (★) pushes the frontier outward past *Instruct* across four instruct models.

**One exponent,
two natures**
THE SYNTHESIS

SHARPEN

$\alpha = 4$ (base) concentrates probability mass on latent reasoning paths.

FLATTEN

$\alpha = 0.5$ releases the creativity suppressed by alignment.

PRINCIPLED

Length-aware matching — no reward model, no correctness verifier, no labels.